

Highlights

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ABSTRACT

Here goes the abstract

1. Introduction

PM_{2.5} is among the most common air pollutants, mainly caused by continuous and intensive emissions from industrial and transportation activities. With the increasing trend of industrialization, its concentration is becoming more severe and widespread. For example, according to a report by Kumar, Raj, Rushwood and Gierens (2025), none of the cities in India met the recommended annual PM_{2.5} concentration of below 5 $\mu\text{g}/\text{m}^3$. Meanwhile, in Europe, it was estimated that 93.9% of registered monitoring stations did not meet this threshold, leading the EU to set a target that is five times higher, at 25 $\mu\text{g}/\text{m}^3$ for 2030. However, around one-third of the stations still exceed this level European Environment Agency (2026). In the U.S., the 90th percentile PM_{2.5} concentration among 345 monitoring sites remained around 9–10 $\mu\text{g}/\text{m}^3$ throughout the 2020–2024 period US Environmental Protection Agency (2026). In China, despite the tangible results of efforts to improve the ecological system, the annual PM_{2.5} concentration remained as high as 28 $\mu\text{g}/\text{m}^3$ in 2025 Xinhua (2026). Consequently, high PM_{2.5} concentrations continue to pose serious threats to human health and the environment.

PM_{2.5} has been associated with numerous chronic diseases. For instance, cardiovascular diseases such as heart attacks and strokes have been linked to PM_{2.5} through mechanisms including cell death in heart cells and disrupted lipid metabolism Yuan, Song, Wang, Xiao, Jia, Lin, Jia, Wang and Ding (2025); Zhang, Zhang, Wu, Nan, Huang, Du, Tian, Liu, Xin, Li, Duan, Chen, Sun and Shen (2021). It also contributes to chronic lung diseases and lung cancer through oxidative stress and inflammation Zhang, Ahn, Wang and Mao (2025); Zhang and Tang (2023). In Europe, around 400,000 premature deaths each year are attributed to ambient air pollution Sattari, Hooyberghs, Janssen, Struzewska, Gawuc, Blyth and Vranckx (2025). Beyond health impacts, high PM_{2.5} concentrations also degrade the overall travel experience in urban areas, reduce living standards, affect real estate values, and increase healthcare costs Sun and Yang (2020).

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Due to these harmful impacts, governmental and international organizations have made numerous attempts to minimize its effects. Among these efforts, forecasting can serve as a foundational basis for numerous actionable solutions, including infrastructure planning, traffic management, resource allocation and as it provides a foreseeable estimation of likely future conditions, allowing for the simulation of potential chains of effects and the testing and evaluation of implemented actions. The basis of forecasting is to produce an estimation of a certain value at a given point in time based on historical records, along with other exogenous variables if provided. When a sufficiently long period of observations is available, PM_{2.5} forecasting can be treated as a regression problem due to its recognized temporal evolution, which is commonly observed in real-world data through cyclical and periodic patterns. As such, in academic research, machine learning (ML) has become a powerful and versatile tool for regression studies, as it allows the discovery of autocorrelation and nonlinear relationships between variables, while also enabling specialized architectures to fit the nature and distinct characteristics of the problem and to serve different purposes, including improving forecasting accuracy, offering real-time predictions, and minimizing the noise caused by outliers.

2. Review methodology

2.1. Data collection

The bibliographic dataset was retrieved from the Scopus database on 9 October 2026. Scopus was chosen for its broad coverage of environmental-science and engineering journals and because it exports complete records, including cited references, without the 1,000-record cap of other platforms, so the whole result set could be analysed rather than a relevance-ranked subset. The search combined three blocks of terms in the title, abstract and keyword fields: the pollutant, the forecasting task and the data-driven modelling family:

```
TITLE-ABS-KEY("PM2.5" OR "PM 2.5" OR "fine particulate
matter")
AND TITLE-ABS-KEY(forecast* OR predict*)
AND TITLE-ABS-KEY("deep learning" OR "machine
learning" OR "neural network*" OR LSTM OR transformer*
OR "graph neural" OR spatiotemporal OR "spatio-temporal")
AND PUBYEAR > 2014 AND PUBYEAR < 2027
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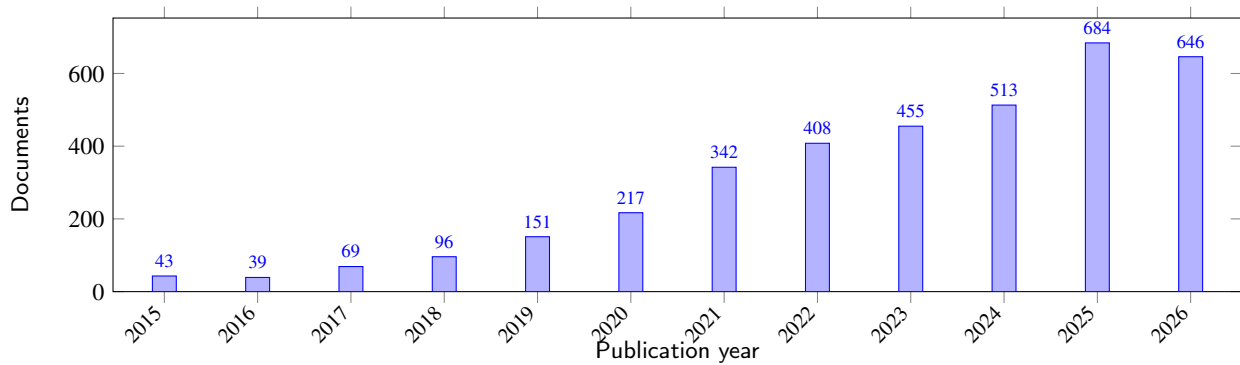


Figure 1: Annual number of PM_{2.5} forecasting and prediction publications in the Scopus corpus (2026 covers January to 9 October).

AND (LIMIT-TO(DOCTYPE, "ar") OR LIMIT-TO(DOCTYPE, "re"))

The time window 2015–2026 covers the period in which deep learning entered air-quality research, and the document-type filter keeps peer-reviewed journal articles and reviews. The query returned 3,663 records (3,605 articles and 58 reviews) from 835 sources and 126 countries, which together had received 95,770 citations at the time of retrieval (median 9 per paper). Most are written in English; 167 are in Chinese with English abstracts and keywords.

Annual output has grown almost every year (Figure 1), from 43 papers in 2015 to 684 in 2025, and the first nine months of 2026 already account for 646. In all, 83.2 % of the corpus was published from 2021 onwards and 50.3 % from 2024 onwards. The field is therefore young and still accelerating, and the most recent methodological developments are well represented in the dataset.

Because the query includes *predict**, the corpus also contains studies that *estimate* PM_{2.5} in space, for example satellite-based exposure mapping, rather than forecast it ahead in time. These were kept on purpose: they share data sources, models and venues with forecasting research, and the bibliometric maps below separate them clearly from forecasting studies. VOSviewer 1.6.20 van Eck and Waltman (2010) was then used to map the intellectual structure of the field through keyword co-occurrence and reference co-citation networks, and the Scopus records were used to profile the leading countries and journals.

2.2. Keyword analysis

Keyword co-occurrence analysis was used to identify the main research themes and the relationships among forecasting methods. Two keywords co-occur when they appear together in the same publication; the more often they do, the stronger their conceptual link. The analysis used author keywords with full counting. Before mapping, a thesaurus of 791 entries merged spelling and notation variants of the same concept (for example “PM_{2.5}”, “pm 2.5” and “fine particulate matter”, or “long short-term memory” and “LSTM”), so that each concept is counted once. To keep the map focused on established concepts, the minimum

number of occurrences was set to 20, which retained 80 keywords. Besides occurrence frequency (Occ.), total link strength (TLS), the sum of a keyword’s co-occurrences with all other retained keywords, was used to gauge how widely each concept is connected across the field. Table 1 lists the keywords by cluster with their TLS, occurrences and average publication year; the VOS clustering technique grouped them into five clusters.

Table 1: Keyword co-occurrence clusters (author keywords, minimum 20 occurrences). TLS: total link strength; Occ.: occurrences; Year: average publication year.

Term	TLS	Occ.	Year
Cluster 1 (red): deep-learning PM_{2.5} forecasting			
deep learning	907	420	2023.44
LSTM	666	279	2023.00
PM _{2.5} prediction	235	139	2023.53
air quality prediction	173	101	2023.86
CNN	245	97	2023.29
PM _{2.5} concentration	89	74	2022.74
PM _{2.5} forecasting	112	63	2023.46
PM _{2.5} concentration prediction	75	61	2023.20
air quality forecasting	117	57	2023.44
RNN	140	55	2022.02
attention mechanism	113	50	2023.70
GRU	124	46	2023.00
time series	91	42	2022.95
ensemble learning	95	34	2023.47
time series forecasting	67	33	2024.27
transformer	80	33	2024.79
air pollution prediction	56	31	2023.55
GNN	63	31	2024.10
hybrid model	74	31	2023.10
SVR	74	31	2022.52
deep neural network	62	27	2023.07
time series analysis	61	27	2023.70
BP neural network	19	23	2020.65
particle swarm optimization	40	23	2022.74
spatiotemporal modeling	31	23	2024.04
spatiotemporal correlation	47	22	2022.18
air pollution forecasting	45	20	2022.95
transfer learning	42	20	2024.15
variational mode decomposition	32	20	2023.75
Cluster 2 (green): PM_{2.5} estimation and remote sensing			
PM _{2.5}	1,533	853	2022.79
machine learning	1,388	759	2023.75
air pollution	1,121	614	2022.99
particulate matter	580	291	2022.71
random forest	475	208	2023.26
aerosol optical depth	288	151	2021.76
ozone	154	74	2022.84
remote sensing	136	60	2023.15
land use regression	85	43	2020.95
China	70	40	2022.62
exposure assessment	58	35	2022.43
prediction model	65	31	2022.84
WRF-Chem	41	31	2022.00
MODIS	67	29	2020.97
aerosol	58	28	2022.64
CMAQ	51	27	2021.89
data fusion	49	25	2022.56
nitrogen dioxide	61	23	2023.09
meteorology	48	22	2023.45
spatiotemporal variation	41	21	2022.95

Term	TLS	Occ.	Year
Cluster 3 (blue): interpretable machine learning for air-quality indices			
prediction	282	118	2022.65
air quality index	237	108	2023.94
XGBoost	185	87	2023.93
air pollutants	94	70	2023.16
SHAP	125	60	2024.90
COVID-19	123	54	2022.39
meteorological factors	104	50	2023.46
explainable AI	71	39	2024.82
feature selection	58	29	2023.10
urban air quality	60	28	2024.21
machine learning models	40	25	2024.76
principal component analysis	47	24	2021.12
correlation analysis	53	22	2023.59
sensitivity analysis	35	20	2023.55
Cluster 4 (yellow): classical machine learning and multiple pollutants			
air quality	576	283	2023.40
ANN	259	135	2022.01
PM ₁₀	206	69	2023.01
SVM	69	35	2022.46
NO ₂	97	29	2023.00
regression	81	27	2022.96
public health	51	26	2023.65
predictive model	59	25	2023.64
pollution	41	22	2022.91
forecast	56	20	2022.35
Cluster 5 (purple): sensing, IoT and forecasting			
neural network	203	94	2021.52
forecasting	189	77	2023.48
artificial intelligence	122	58	2023.53
indoor air quality	95	56	2023.50
low-cost sensors	118	55	2024.15
internet of things	128	51	2023.61
air quality monitoring	52	34	2024.41

2.2.1. Network visualisation

Figure 2 shows the structure of the keyword network. Node size is proportional to occurrences, link thickness to co-occurrence strength, and keywords that frequently appear together sit close to one another. The five clusters form two large poles, connected by three smaller, more application-oriented groups.

The *red cluster* (Cluster 1, 29 keywords) is the deep-learning forecasting core. It is anchored by *deep learning* (420 occurrences) and *LSTM* (279), followed by *CNN*, *RNN* and *GRU*, and by the task terms *PM_{2.5} prediction*, *PM_{2.5} forecasting* and *air quality prediction*. Around this core sit the mechanisms used to improve recurrent and convolutional forecasters: *attention mechanism*, *hybrid model*, signal decomposition (*variational mode decomposition*), metaheuristic tuning (*particle swarm optimization*), *transfer learning*, and, at the edge of the cluster, *transformer* and *GNN* together with *spatiotemporal modeling* and *spatiotemporal correlation*. This is where multi-station, multi-step forecasting research such as the present study is located.

The *green cluster* (Cluster 2, 20 keywords) is the second pole and contains the most frequent keywords overall: *PM_{2.5}* (853), *machine learning* (759) and *air pollution* (614). Its specific vocabulary, *aerosol optical depth*, *MODIS*, *remote sensing*, *land use regression*, *data fusion*, *exposure assessment* and the chemical transport models *WRF-Chem* and *CMAQ*, identifies it as the spatial estimation and exposure-mapping stream. In this stream, tree ensembles such as *random forest* (208) predominate. Its high TLS reflects the many generic keywords it contains rather than closeness to the forecasting core: on the map it sits on the opposite side from the red cluster.

The *blue cluster* (Cluster 3, 14 keywords) gathers interpretable machine learning for air-quality indices: *air quality index*, *XGBoost*, *feature selection*, *meteorological factors*, *SHAP* and *explainable AI*, together with *COVID-19*, reflecting the many studies of lockdown effects on pollution. The *yellow cluster* (Cluster 4, 10 keywords) groups earlier shallow learners (*ANN*, *SVM*, *regression*) applied to several pollutants (*PM₁₀*, *NO₂*) and to *public health*. The *purple cluster* (Cluster 5, 7 keywords) links forecasting with sensing infrastructure: *internet of things*, *low-cost sensors*, *indoor air quality* and *air quality monitoring*.

2.2.2. Overlay visualisation

Figure 3 adds time: each keyword is coloured by the average publication year of the papers that use it, from blue/purple (around 2022 or earlier) through green to yellow (2024 or later). The oldest themes are the spatial-estimation and shallow-learning vocabulary: *BP neural network* (2020.65), *land use regression* (2020.95), *MODIS* (2020.97), *principal component analysis* (2021.12) and *aerosol optical depth* (2021.76). Recurrent networks occupy the middle of the period (*RNN* 2022.02, *LSTM* 2023.00, *GRU* 2023.00). The most recent themes, all with average years from 2024, are *SHAP* (2024.90), *explainable AI* (2024.82), *transformer* (2024.79), *air quality monitoring*

(2024.41), *time series forecasting* (2024.27), *transfer learning* (2024.15), *low-cost sensors* (2024.15), *GNN* (2024.10) and *spatiotemporal modeling* (2024.04).

Three shifts follow from this timeline. First, the field has moved from estimating *PM_{2.5}* in space towards forecasting it in time, and from shallow networks towards deep sequence models. Second, within deep forecasting, attention-based and graph-based architectures, which model dependencies across time steps and across stations, are the newest and fastest-growing topics, but they are still small (33 and 31 occurrences respectively). Third, interpretability (*SHAP*, *explainable AI*) and dense low-cost sensing have become active concerns in parallel with model development. As in other forecasting domains, the change is an expansion rather than a replacement: *LSTM* and *CNN* models remain the most used deep architectures, now increasingly combined with attention, decomposition and graph components.

2.2.3. Density visualisation

The density visualisation in Figure 4 shows where research attention is concentrated. The brightest region surrounds *PM_{2.5}*, *machine learning* and *air pollution*, the shared vocabulary of the whole corpus, with a second dense area around *deep learning* and *LSTM*. Architecture-specific terms such as *transformer*, *GNN* and *spatiotemporal modeling* lie in low-density areas at the edge of the deep-learning region, which confirms that, despite their recent growth, multi-station spatio-temporal forecasting architectures remain a comparatively small part of the literature.

2.3. Co-citation analysis

Co-citation analysis maps the field's intellectual foundations: two references are co-cited when a later paper cites both, so clusters of co-cited works show the bodies of knowledge that research builds on. Cited references were first standardised, because the same work is often cited in slightly different forms (author initials, journal abbreviations, page formats); a thesaurus of 4,762 entries merged these variants. With a minimum of 40 citations, 101 references were retained and grouped into three clusters (Figure 5; Table 2 lists the four most cited works in each).

The *red cluster* (46 references) is the deep-learning forecasting foundation. It is headed by the original *LSTM* paper Hochreiter and Schmidhuber (1997b), followed by the first deep forecasting models for *PM_{2.5}*: *LSTM*-based predictors Li et al. (2017), the *CNN-LSTM* hybrid of Huang and Kuo Huang and Kuo (2018) and the graph-convolution-*LSTM* model of Qi et al. Qi et al. (2019), an early example of combining station graphs with recurrent networks. The *green cluster* (35 references) is the satellite-based estimation literature, e.g. Ma et al. (2014); Di et al. (2016); Hu et al. (2017); Chen et al. (2018), which maps *PM_{2.5}* from aerosol optical depth, meteorology and land use, mostly over China and the United States. The *blue cluster* (20 references) joins the machine-learning methods shared by both streams, random forests Breiman (2001) and gradient boosting Chen and Guestrin (2016), with the health-burden

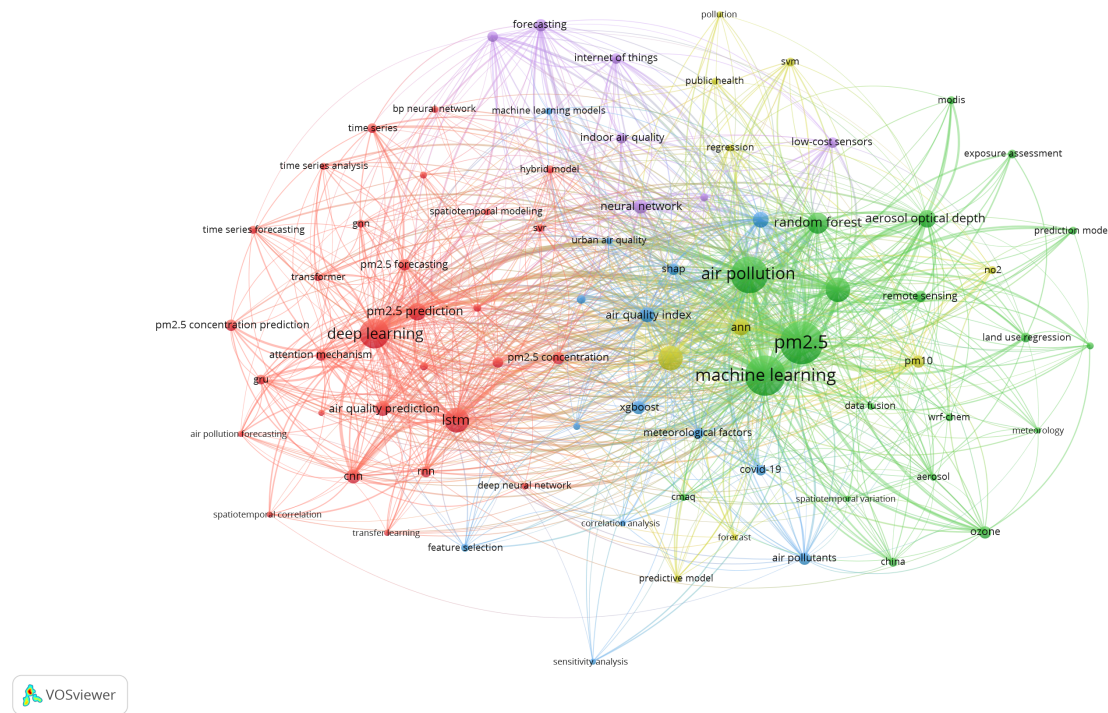


Figure 2: Keyword co-occurrence network of PM_{2.5} forecasting and prediction research, 2015–2026 (author keywords, minimum 20 occurrences, 80 keywords, five clusters).

Table 2
Most co-cited references in each co-citation cluster.

Reference	Citations	TLS
Cluster 1 – Red: deep-learning forecasting		
Hochreiter & Schmidhuber (1997) Hochreiter and Schmidhuber (1997b)	330	963
Huang & Kuo (2018) Huang and Kuo (2018)	163	661
Qi et al. (2019) Qi, Li, Karimian and Liu (2019)	158	729
Li et al. (2017) Li, Peng, Yao, Cui, Hu, You and Chi (2017)	145	625
Cluster 2 – Green: satellite-based estimation		
Hu et al. (2017) Hu, Belle, Meng, Wildani, Waller, Strickland and Liu (2017)	120	719
Di et al. (2016) Di, Kloog, Koutrakis, Lyapustin, Wang and Schwartz (2016)	97	601
Ma et al. (2014) Ma, Hu, Huang, Bi and Liu (2014)	93	687
Chen et al. (2018) Chen, Li, Knibbs, Hamm, Cao, Li, Guo, Ren, Abramson and Guo (2018)	83	447
Cluster 3 – Blue: machine-learning methods and health burden		
Breiman et al. (2001) Breiman (2001)	357	1,052
Chen & Guestrin (2016) Chen and Guestrin (2016)	131	401
Lelieveld et al. (2015) Lelieveld, Evans, Fnais, Giannadaki and Pozzer (2015)	121	331
Cohen et al. (2017) Cohen, Brauer, Burnett et al. (2017)	101	353

studies Lelieveld et al. (2015); Cohen et al. (2017) and air-quality guidelines that motivate the research. The forecasting (red) and estimation (green) foundations are linked mainly through this blue methodological bridge, which matches the two-pole structure of the keyword map.

2.4. Country-level contributions and research impact

The country of the authors' affiliations shows where research capacity and funding are concentrated. Countries were counted with full counting, so a paper counts once for each country among its affiliations; total link strength is the number of country co-authorship links, a measure

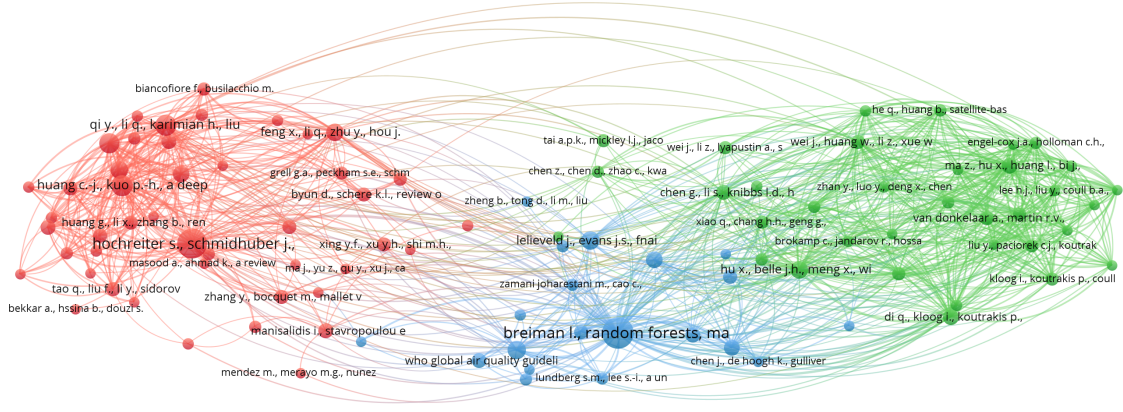


Figure 5: Co-citation network of cited references (minimum 40 citations, 101 references, three clusters).

a test RMSE of $4.4 \mu\text{g m}^{-3}$ at a 1 h lead, rising to about $11.7 \mu\text{g m}^{-3}$ beyond 12 h.

4. Methodology

4.1. Problem Statement

Given the data with size $C \times L$, where C is the number of channels and L is the length of the time series, the series consists of observations $\{x_1, x_2, \dots, x_L\}$, where $\{1, 2, \dots, L\}$ denotes the time indices. At a specific time point t , the observation of channel c is denoted as $x_t^{(c)}$.

In our problem, C consists of different meteorological channels, as described in Subsection ?? and PM2.5 records. The sequence length L is further divided into the historical length H and the forecast horizon F , which signify the historical input length and forecasted length, respectively. Here, H determines how many previous time steps are taken as input for the model, while F specifies how many future time steps are predicted.

In general, this study aims to forecast $\text{PM}_{2.5}$ based on its own historical observations and other meteorological information. Thus, we have C input channels and one output channel. The forecasting problem can be formulated as taking an input sequence $\mathbf{X} \in \mathbb{R}^{C \times H}$ to predict the forecasted sequence $\hat{\mathbf{Y}} \in \mathbb{R}^{1 \times F}$. The prediction error is computed by taking the difference between the ground truth sequence $\mathbf{Y} \in \mathbb{R}^{1 \times F}$ and the forecasted sequence $\hat{\mathbf{Y}} \in \mathbb{R}^{1 \times F}$, where $\mathbf{Y}$ denotes the $\text{PM}_{2.5}$ concentration sequence.

4.2. Temporal Encoding

Temporal development is considered the backbone for projecting future forecasts, as it preserves historical evolving behavior and thus plays a major role in predicting future time steps given the inherent temporal patterns. The GRU (Gated Recurrent Unit) is applied at this stage because it belongs to the recurrent neural network family and is known for efficiently capturing long-term dependencies, thanks to its simplified structure compared to its predecessor, the LSTM (Long Short-Term Memory). In specific, there are mainly 2 modules in this architecture, called reset gate and update gate and is controlled by the following equations:

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t] + b_r) \text{ (reset gate)}, \quad (1)$$

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t] + b_z) \text{ (update gate)}, \quad (2)$$

$$h'_t = \tanh(W_h \cdot [r_t \odot h_{t-1}, x_t] + b_h), \quad (3)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot h'_t. \quad (4)$$

While the reset gate and update gate possess identical mathematical operations that weight the previous hidden state and the current input at time step t , they have distinct roles. The reset gate directly enters the candidate hidden state computation, guiding how much of the prior hidden state is allowed to contribute, together with the current input, to form the candidate state. Meanwhile, the update gate provides the weights that balance the prior hidden state and the candidate state when forming the present hidden state. This continuous update mechanism, by balancing past memory and present input, allows the GRU to inject historical

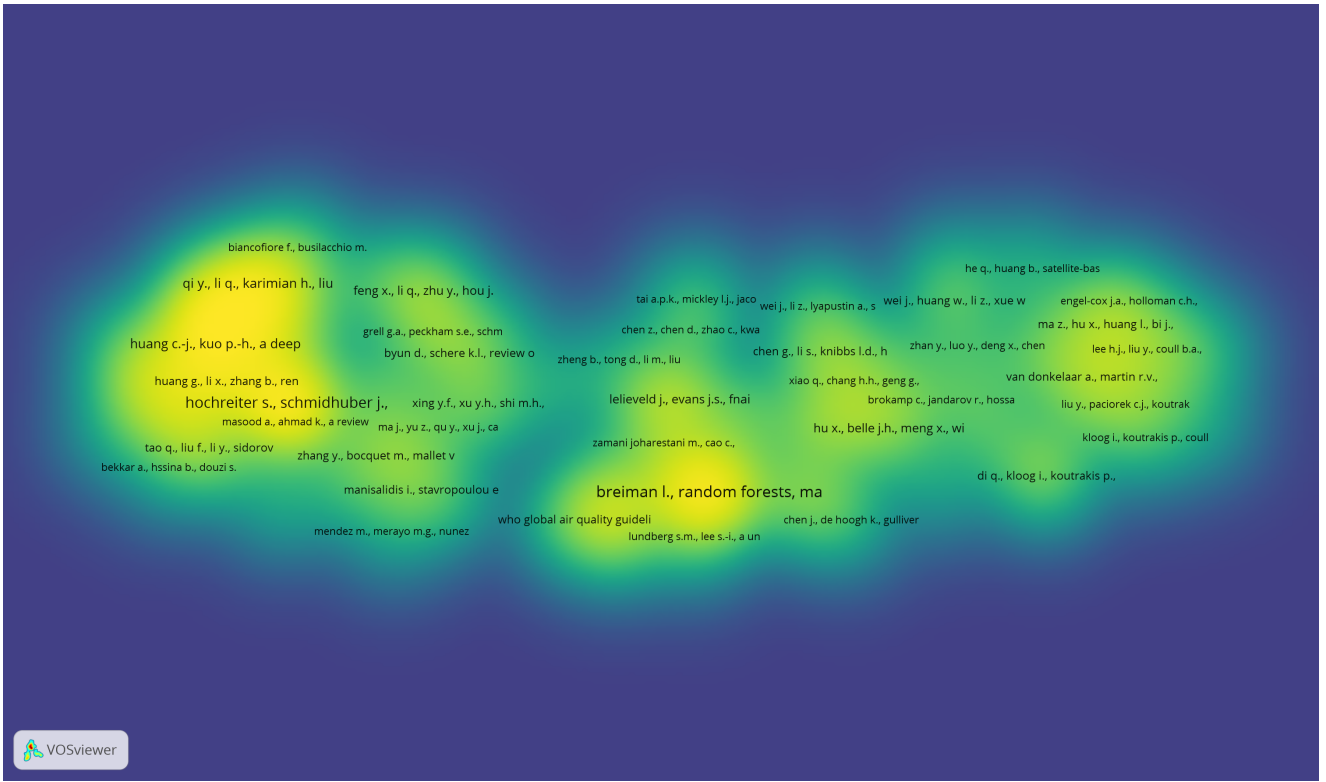


Figure 6: Density visualisation of the co-citation network.

information together with the latest observations, enabling predictions that are effectively grounded in both long-term context and current dynamics. At our current problem, at each station their hidden states will be produced, then further feed into the nt stage for intergrating temporal and spatial together.

4.3. Joint Spatio-Temporal Encoding

The current station does not have its $PM_{2.5}$ concentration determined only by its own emissions, but also by neighboring sources. Therefore, rather than considering each station's historical records independently through the GRU, embedding spatial context between stations is an ideal step. Specifically, at this stage, an attention mechanism is employed to generate between-station interactions based on their hidden representations, allowing neighboring stations to exchange information while their temporal representations are still being formed.

For each timestep, the hidden representations are projected into query, key, and value vectors as

$$Q = W_Q H, \quad K = W_K H, \quad V = W_V H, \quad (5)$$

where H denotes the hidden representations produced by the GRU.

To make this module more adaptive to a real atmospheric transport problem, where stations that are farther apart generally have weaker interactions, a distance-based term is added to the attention score before the softmax

operation. The attention score between station i and station j is computed as

$$s_{ij} = \frac{Q_i K_j^T}{\sqrt{d}} + w_d \exp\left(-\frac{d_{ij}}{\sigma_d}\right), \quad (6)$$

where d_{ij} is the Haversine distance between the two stations, σ_d controls the distance decay, and w_d is a learnable weight. The attention weights are then obtained by

$$\alpha_{ij} = \text{softmax}(s_{ij}). \quad (7)$$

Finally, the aggregated neighboring information is added back to the original hidden representation through a residual connection, followed by Layer Normalization,

$$\tilde{H} = \text{LayerNorm}(H + \alpha V), \quad (8)$$

which stabilizes training while allowing temporal and spatial information to be learned jointly.

4.4. Context-Adaptive Diffusivity and Anisotropic Transport

Relying only on learned feature interactions to estimate the influence between stations may lose the actual way that emissions are transported in the atmosphere, where transport is largely governed by physical and natural factors such as

Table 3

Top 20 countries by number of documents (full counting; Hong Kong included in China). TLS: total link strength of country co-authorship.

No.	Country	Docs	Share (%)	Citations	Cit./doc	TLS
1	China	1,709	46.7	51,205	30.0	670
2	United States	695	19.0	29,436	42.4	616
3	India	426	11.6	7,633	17.9	178
4	South Korea	219	6.0	3,933	18.0	118
5	United Kingdom	157	4.3	5,771	36.8	301
6	Taiwan	148	4.0	5,783	39.1	96
7	Iran	100	2.7	2,092	20.9	52
8	Australia	94	2.6	4,229	45.0	172
9	Canada	82	2.2	2,849	34.7	130
10	Thailand	78	2.1	609	7.8	50
11	Saudi Arabia	76	2.1	1,227	16.1	175
12	Malaysia	75	2.0	1,381	18.4	108
13	Italy	67	1.8	2,934	43.8	143
14	Germany	65	1.8	2,201	33.9	161
15	Indonesia	52	1.4	688	13.2	41
16	Pakistan	47	1.3	616	13.1	65
17	Turkey	46	1.3	781	17.0	51
18	Japan	45	1.2	1,037	23.0	44
19	Israel	40	1.1	3,616	90.4	64
20	France	38	1.0	1,497	39.4	108

Table 4

Top 10 journals by number of documents.

No.	Journal	Docs	Citations	Cit./doc
1	Atmospheric Environment	175	7,212	41.2
2	Atmosphere	162	3,172	19.6
3	Science of the Total Environment	120	7,081	59.0
4	Atmospheric Pollution Research	105	3,573	34.0
5	Environmental Pollution	103	4,759	46.2
6	Scientific Reports	77	1,835	23.8
7	Environmental Science and Technology	66	4,971	75.3
8	Environmental Research	66	2,504	37.9
9	Sustainability (Switzerland)	65	1,251	19.2
10	Environment International	63	4,035	64.0

wind speed and wind direction. Intuitively, wind carries $PM_{2.5}$ along its path, allowing pollutants emitted at one station to reach another. Therefore, simulating this transport process is worth exploring.

Let i denote the receiver (target station) and j denote the source station. Let θ_{ij} be the angle between the source

station's wind direction and the bearing from station j to station i . Instead of treating the source–receiver relationship as a single radial distance, the displacement is decomposed into downwind and crosswind coordinates,

Table 5

Features per station and hour. Statistics are over all station-hours; for $PM_{2.5}$, over measured values only.

Feature	Unit	Source	Mean $\pm$ SD	Range
Temperature (2 m)	°C	ERA5	10.2 $\pm$ 8.5	−38.8 – 42.8
Relative humidity (2 m)	%	ERA5	73.3 $\pm$ 20.4	2 – 100
Precipitation	mm h ^{−1}	ERA5	0.15 $\pm$ 0.44	0 – 10.7
Wind speed (10 m)	m s ^{−1}	ERA5	2.1 $\pm$ 1.3	0 – 16.7
Wind direction (10 m)	°	ERA5	–	0 – 360
$PM_{2.5}$ (target)	$\mu\text{g m}^{-3}$	AirNow	6.0 $\pm$ 8.8	0 – 687

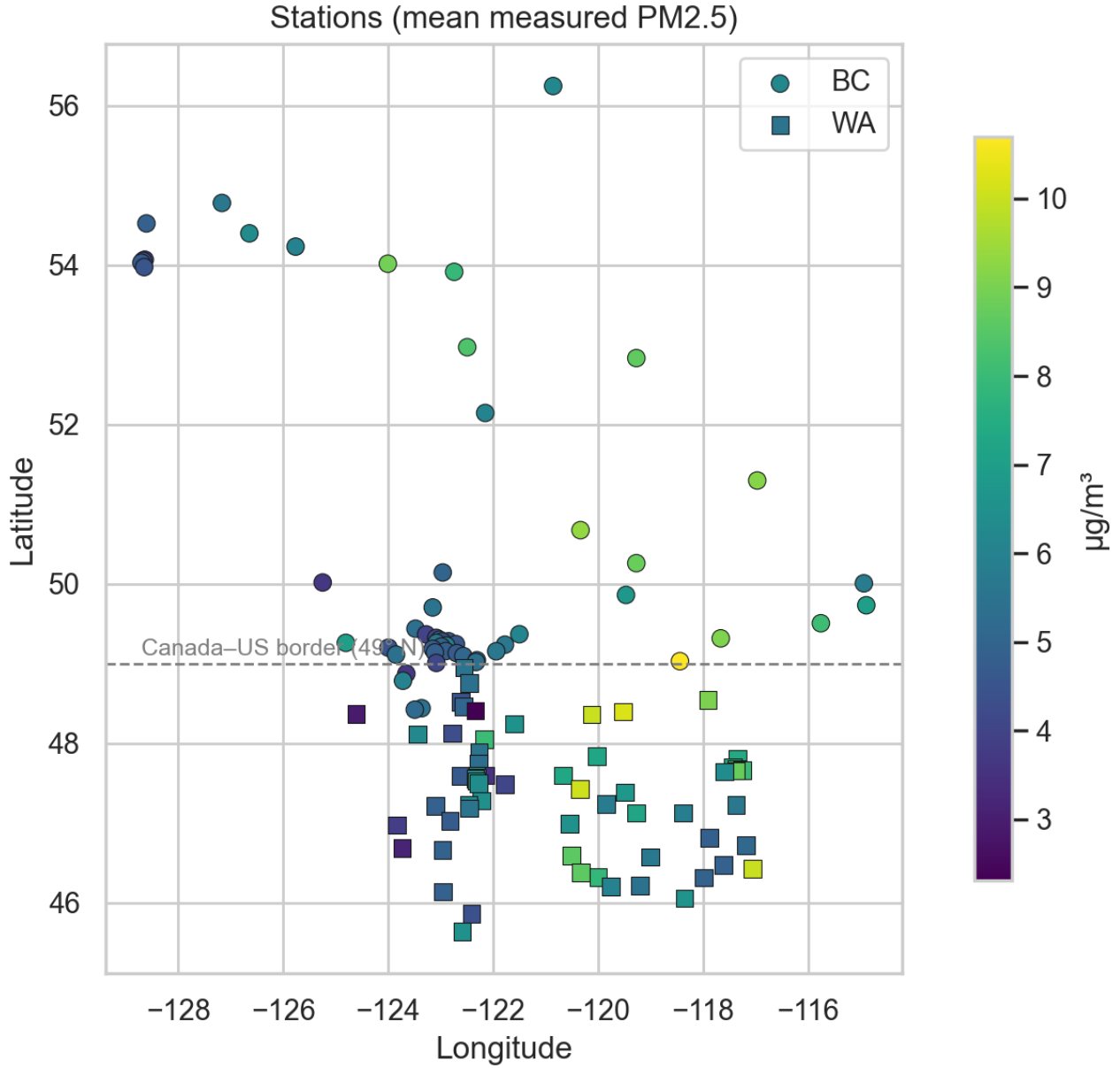


Figure 7: The 115 monitoring stations (circles: BC; squares: WA), coloured by mean measured $PM_{2.5}$. The dashed line marks the Canada–US border ($49^\circ N$).

$$x_{ij} = d_{ij} \cos \theta_{ij}, \quad y_{ij} = d_{ij} \sin \theta_{ij}, \quad (9)$$

where d_{ij} is the Haversine distance between the two stations. The downwind component represents transport along the wind direction, while the crosswind component represents dispersion perpendicular to it.

In general, the objective is to estimate how much $PM_{2.5}$ emitted from source station j can reach receiver station i . Classical atmospheric transport can be described by the advection–diffusion equation,

$$\frac{\partial C}{\partial t} = D \nabla^2 C - \mathbf{v} \cdot \nabla C, \quad (10)$$

where C denotes pollutant concentration, $\mathbf{v}$ is the wind velocity, and D is the diffusion coefficient. A larger diffusion coefficient results in faster spreading of pollutants, while a smaller value keeps the plume more concentrated.

However, using a single global diffusion coefficient is unrealistic because atmospheric mixing varies across locations and time. In the proposed model, diffusivity is treated as a property of the source station's local environment at each historical lag. The source context is represented as

$$c_{jk} = \left[\frac{s_{jk}}{10}, \sin \omega_{jk}, \cos \omega_{jk}, \frac{e_j}{1000}, p_{jk} \right], \quad (11)$$

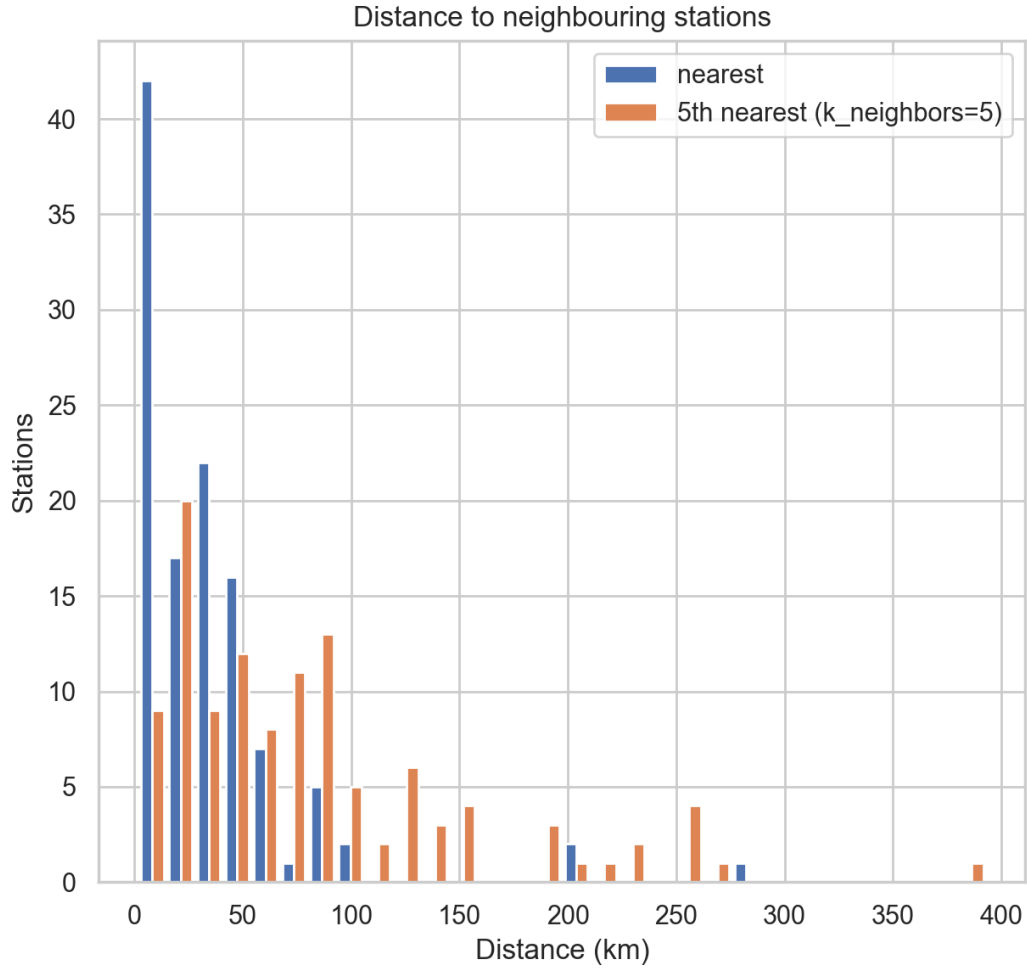


Figure 8: Distance from each station to its nearest and 5th-nearest neighbour. The graph links each station to its $k = 5$ nearest neighbours.

Table 6
Monitoring network.

Region	Operating agency	Stations
BC, Canada	BC Ministry of Environment	34
	Metro Vancouver	20
WA, USA	Washington Department of Ecology	60
	Spokane Regional Clean Air Agency	1
Total		115
Extent		45.6–56.2°N, 128.7–114.9°W
Nearest-station distance, median (range)		26.5 (1.7–286) km
5th-nearest-station distance, median		64 km
Data completeness, median (min)		97% (85%)
Missing station-hours (gap-filled)		4.0%
Elevation, median (range)		104 (0–1,335) m

where s_{jk} is wind speed, ω_{jk} is the hour of day, e_j is station elevation, and p_{jk} is the recent $\text{PM}_{2.5}$ concentration at the source station.

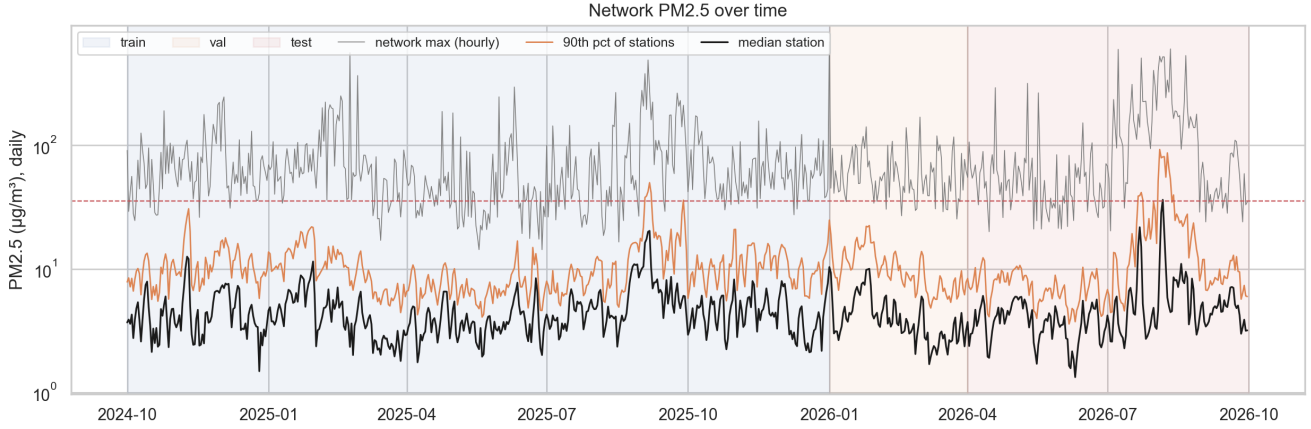
This context vector is passed through a two-layer multi-layer perceptron with a Linear–Tanh–Linear architecture,

$$u_{jk} = \tanh(W_1 c_{jk} + b_1), \quad (12)$$

$$z_{jk} = W_2 u_{jk} + b_2, \quad (13)$$

Table 7Chronological splits and statistics of measured $\text{PM}_{2.5}$.

Split	Period	Hours	Median ($\mu\text{g m}^{-3}$)	99th pct. ($\mu\text{g m}^{-3}$)	$>35.5 \mu\text{g m}^{-3}$ (% hours)
Train	Oct 2024 – Dec 2025	10,968	4.1	30.0	0.6
Validation	Jan 2026 – Mar 2026	2,160	3.8	26.6	0.4
Test	Apr 2026 – Sep 2026	4,392	4.0	64.4	2.5

**Figure 9:** Daily $\text{PM}_{2.5}$ across the network: median station (black), 90th percentile of stations (orange) and network maximum (grey), with the train, validation and test periods shaded. The dashed line marks $35.5 \mu\text{g m}^{-3}$.

where $z_{jk} = [z_{jk}^{\parallel}, z_{jk}^{\perp}]$ are the raw outputs. Finally, a Softplus activation is applied to ensure physically valid positive diffusivities,

$$D_{jk}^{\parallel} = \text{softplus}(z_{jk}^{\parallel}), \quad D_{jk}^{\perp} = \text{softplus}(z_{jk}^{\perp}), \quad (14)$$

where $D^{\parallel}$ and $D^{\perp}$ represent the along-wind and cross-wind diffusion coefficients, respectively. These adaptive diffusivities are then used in the two-dimensional anisotropic Green's function to estimate the transport contribution from each source station to each receiver station.

4.4.1. Anisotropic Green's Function

After obtaining the surrounding context and adaptive diffusivities, one of the core questions is how much $\text{PM}_{2.5}$ from source station j at historical lag k can reach receiver station i after the corresponding elapsed time. This transport is estimated using the Green's function of the advection–diffusion equation.

The classical advection–diffusion equation is

$$\frac{\partial C}{\partial t} = D \nabla^2 C - \mathbf{v} \cdot \nabla C, \quad (15)$$

where C denotes pollutant concentration, $\mathbf{v}$ is the wind velocity, and D is the diffusion coefficient.

Although this formulation describes atmospheric transport, the simpler radial formulation still treats the source–receiver relationship using a single distance and a single

diffusion coefficient. Consequently, it cannot distinguish whether two stations at the same distance are located directly downwind or crosswind of the source.

To better represent this process, the source–receiver displacement is projected onto a wind-aligned coordinate system,

$$x_{ij} = d_{ij} \cos \theta_{ij}, \quad y_{ij} = d_{ij} \sin \theta_{ij}, \quad (16)$$

where d_{ij} is the Haversine distance between stations and θ_{ij} is the angle between the source station's wind direction and the bearing from station j to station i . Here, x_{ij} represents the downwind displacement, while y_{ij} represents the crosswind displacement.

Instead of using a single diffusion coefficient, the adaptive coefficients learned in the previous subsection, $D_{jk}^{\parallel}$ and $D_{jk}^{\perp}$, are directly incorporated into the transport kernel. The resulting anisotropic Green's function becomes

$$G_{ijk} = \frac{1}{4\pi \sqrt{D_{jk}^{\parallel} D_{jk}^{\perp}} t_k} \exp \left(-\frac{(x_{ij} - v_{jk} t_k)^2}{4 D_{jk}^{\parallel} t_k} - \frac{y_{ij}^2}{4 D_{jk}^{\perp} t_k} \right). \quad (17)$$

For the downwind direction, the term $(x_{ij} - v_{jk} t_k)^2$ follows a simple physical intuition. The plume reaches its maximum contribution when the distance travelled by the wind, $v_{jk} t_k$, matches the downwind distance x_{ij} , giving

$(x_{ij} - v_{jk}t_k)^2 = 0$. When $x_{ij} > v_{jk}t_k$, the plume has not yet reached the receiver, whereas when $x_{ij} < v_{jk}t_k$, the plume has already passed the receiver. In both cases, the exponential term reduces the transport contribution smoothly.

For the crosswind direction, only the perpendicular displacement appears,

$$\exp\left(-\frac{y_{ij}^2}{4D_{jk}^\perp t_k}\right). \quad (18)$$

Unlike the downwind term, there is no advection component because the wind does not transport pollutants perpendicular to its direction. Instead, dispersion in this direction is governed solely by turbulent diffusion. As the angle between the wind direction and the station bearing increases, y_{ij} becomes larger, leading to a weaker transport contribution for stations located farther from the plume centreline.

4.5. Physics-Fused Attention

Under real-world atmospheric transport, simulating the influence of natural factors is important. Therefore, the attention logit A_{ijk} is designed to capture the interaction between receiver station i and source station j at historical lag k by incorporating both learned feature similarity and physically meaningful factors, including terrain similarity, wind alignment, geographical distance, and travel-time consistency. In addition to these hand-crafted physical terms, a dynamic Green's function term is also injected into the attention score, allowing the transport estimate derived from the advection–diffusion process to directly influence the attention mechanism.

The attention logit is computed as

$$A_{ijk} = A_{ijk}^{\text{content}} + w_{\text{dist}} B_{ij}^{\text{dist}} + w_{\text{wind}} B_{ijk}^{\text{wind}} + w_{\text{terrain}} B_{ij}^{\text{terrain}} + w_{\text{lag}} B_{ijk}^{\text{lag}} + w_{\text{phys}} \log(G_{ijk} + \epsilon), \quad (19)$$

where A_{ijk}^{content} is the query–key similarity score, B_{ij}^{dist} is the distance decay term, B_{ijk}^{wind} represents wind alignment, B_{ij}^{terrain} measures terrain similarity, B_{ijk}^{lag} models travel-time consistency, and G_{ijk} is the anisotropic Green's function introduced in the previous subsection.

Unlike the temporal graph encoder, which performs attention only among stations at the same timestep, this module evaluates every source station across every historical lag for each receiver station. This provides richer spatio-temporal context by allowing the model to determine not only which neighboring station is important, but also which historical observation from that station is most relevant. The final attention weights are then obtained by applying the softmax function over all source station–lag pairs,

$$\alpha_{ijk} = \text{softmax}_{(j,k)}(A_{ijk}), \quad (20)$$

which are subsequently used to compute the weighted aggregation of neighboring hidden representations.

4.6. Continuous ODE-Driven Transport

The model components so far explore historical observations through learned parameters and separate processing stages. However, under real atmospheric conditions, the evolution of $\text{PM}_{2.5}$ is a continuous process rather than a sequence of independent transitions. Therefore, a continuous-time transport module is constructed in parallel with the GRU decoder to simulate the ongoing development of pollutant transport from source stations toward the target station.

Unlike the GRU branch, which learns temporal forecasting patterns from historical observations, the ODE branch is initialized only from physical information, including the last observed $\text{PM}_{2.5}$ concentration, wind speed, and wind direction. These variables are projected into a physics latent feature vector,

$$z_i^{\text{phy}}(0) = W_{\text{init}} x_i^{\text{phy}} + b_{\text{init}}, \quad (21)$$

which represents the transport state of each station rather than the pollutant concentration itself.

This parallel branch provides two main benefits. First, because it relies only on physical variables, it offers a dedicated representation of the surrounding transport environment without mixing it with the learned temporal features. Second, instead of treating transport as a one-time estimate from historical observations, it continuously evolves the transport state throughout the prediction horizon. Consequently, the model maintains an ongoing physical view of pollutant movement before injecting this continuously updated transport representation back into the GRU decoder through a learned fusion gate.

Then, to simulate the exchange of transport information between neighboring stations and the target station, the physics latent representation is evolved according to

$$\frac{dz_i^{\text{phy}}}{dt} = \sum_{j \in \mathcal{N}(i)} g_{ij} (z_j^{\text{phy}} - z_i^{\text{phy}}) + \beta_i z_i^{\text{phy}}, \quad (22)$$

where z_i^{phy} is the physics latent feature vector of station i , $\mathcal{N}(i)$ denotes the neighboring stations, g_{ij} is the transport conductance derived from the anisotropic Green's function, and β_i is the learnable local reaction coefficient.

The first term models the exchange of transport information between neighboring stations. The difference $(z_j^{\text{phy}} - z_i^{\text{phy}})$ measures the transport-state gap between two stations, while g_{ij} determines how strongly information is exchanged. When g_{ij} is small, only a limited amount of transport information flows from station j to station i , whereas a larger g_{ij} results in a stronger exchange. The second term, $\beta_i z_i^{\text{phy}}$, captures the local evolution of the transport state at station i , allowing the latent representation to be amplified or attenuated independently of neighboring interactions.

After initializing the physics latent state and constructing the exchange conductance g_{ij} , the transport dynamics are integrated continuously across the entire prediction horizon

using a Neural ODE. Unlike a fixed-step recurrent decoder, the ODE solver evaluates the dynamics at arbitrary intermediate time points selected adaptively during numerical integration. Consequently, future meteorological variables cannot be queried at every solver step.

To preserve a physically consistent transport process, the exchange conductance is computed once from the last observed historical context and remains fixed throughout the integration. Specifically, the last observed wind direction, wind speed, hour of day, and $\text{PM}_{2.5}$ concentration are used to construct g_{ij} before integration begins, after which the ODE evolves autonomously over time.

The latent trajectory is obtained by

$$z^{phy}(t) = \text{odeint}(f, z_0^{phy}, t), \quad (23)$$

where $\text{odeint}(\cdot)$ numerically integrates the transport dynamics from the initial physics latent state across the prediction horizon. The initial condition at $t = 0$ is discarded because it corresponds to the last observed historical state, and only the continuously evolved latent representations for future timesteps are retained for subsequent decoder fusion.

5. Results

Protocol. All models receive 24 h of history and forecast the next 24 h at every station. They are trained on the training split, early-stopped on validation, and evaluated on the test split (April–September 2026; Sec. 3). Only forecast hours are scored, and only at measured (not gap-filled) station-hours. We report RMSE, MAE and MAPE (MAPE excludes true values below $1 \mu\text{g m}^{-3}$), and RMSE on hours whose true value exceeds $35.5 \mu\text{g m}^{-3}$. For high-pollution events we report the critical success index (CSI), probability of detection (POD) and false-alarm ratio (FAR) at the US AQI thresholds of $35.5 \mu\text{g m}^{-3}$ (unhealthy for sensitive groups) and $55.5 \mu\text{g m}^{-3}$ (unhealthy). Each model is trained with five random seeds, and we report mean $\pm$ sample standard deviation. Persistence (repeating the last observed hour over the whole horizon) is included as a reference. Models whose runs have not yet finished are shown as “–”.

Overall accuracy. AirLapseV2 has the lowest RMSE ($9.35 \mu\text{g m}^{-3}$; Table 8), 14.5% below persistence. Its margin over $\text{PM}_{2.5}$ -GNN (1.2%) is significant across seeds (Welch’s t -test, $p = 0.013$). Its margin over the second-best model, STMamba (0.9%), is not ($p = 0.19$). AirLapseV2 also varies least across seeds (std 0.04 vs. 0.06–0.23 for the other models with five seeds). On MAE, $\text{PM}_{2.5}$ -GNN (3.16), Informer (3.18) and AirLapseV2 (3.19) are statistically indistinguishable ($p > 0.19$), and Informer has the lowest MAPE. The three models with the lowest RMSE all model interactions between stations explicitly. The recurrent models, which forecast each station from its own history, trail by about $0.7 \mu\text{g m}^{-3}$. The MLP does worse than persistence.

High-pollution hours. All models are much less accurate during high pollution. RMSE on hours above $35.5 \mu\text{g m}^{-3}$ (49.7 – $69.1 \mu\text{g m}^{-3}$) is more than five times the overall RMSE, which shows that peaks are systematically under-predicted. TCN-DIR, which is trained with an imbalanced-regression objective, has the lowest peak RMSE ($49.7 \mu\text{g m}^{-3}$) and the highest POD (0.600 at $35.5 \mu\text{g m}^{-3}$). It pays for this with the highest FAR (0.456) and an MAE 58% above AirLapseV2’s. AirLapseV2 has the second-lowest peak RMSE ($52.0 \mu\text{g m}^{-3}$), but its event-detection scores are mid-range (CSI 0.391 at $35.5 \mu\text{g m}^{-3}$) and vary more across seeds than $\text{PM}_{2.5}$ -GNN’s (Table 9). Notably, persistence has a higher CSI than every learned model at both thresholds (0.432 and 0.397). Smoke episodes persist for days, so the last observed hour often remains above the threshold. The learned models, by contrast, were trained on a period in which only 0.6% of hours exceeded $35.5 \mu\text{g m}^{-3}$, against 2.5% in the test period, and they shrink their forecasts toward typical concentrations. Reliable exceedance forecasting under this distribution shift remains an open problem for all models, including AirLapseV2.

CRedit authorship contribution statement

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Table 8

Forecast error on the test split (24 h horizon, measured values only; mean $\pm$ std over 5 seeds). Lower is better. **Bold**: best learned model; underlined: second best. [†]Fewer than five seeds completed (AirFormer: 3; AirPhyNet: 1). –: not yet available.

Model	RMSE ($\mu\text{g m}^{-3}$)	MAE ($\mu\text{g m}^{-3}$)	MAPE (%)	RMSE _{>35.5} ($\mu\text{g m}^{-3}$)
Persistence	10.95	3.70	61.4	53.0
MLP	11.71 $\pm$ 0.14	3.90 $\pm$ 0.04	65.3 $\pm$ 1.5	69.1 $\pm$ 1.0
LSTM	10.02 $\pm$ 0.06	3.32 $\pm$ 0.01	52.0 $\pm$ 0.7	57.4 $\pm$ 0.7
GRU	10.02 $\pm$ 0.16	3.33 $\pm$ 0.02	53.5 $\pm$ 1.2	57.5 $\pm$ 1.3
Transformer	–	–	–	–
Informer	9.60 $\pm$ 0.10	<u>3.18</u> $\pm$ 0.02	49.6 $\pm$ 1.4	54.3 $\pm$ 0.9
Autoformer	9.82 $\pm$ 0.08	3.53 $\pm$ 0.03	60.5 $\pm$ 0.9	52.2 $\pm$ 0.5
Crossformer	10.26 $\pm$ 0.13	3.38 $\pm$ 0.03	54.8 $\pm$ 1.3	59.3 $\pm$ 0.9
PM _{2.5} -GNN	9.46 $\pm$ 0.06	3.16 $\pm$ 0.02	51.5 $\pm$ 1.0	53.8 $\pm$ 0.5
AirFormer [†]	9.66 $\pm$ 0.13	3.23 $\pm$ 0.03	51.9 $\pm$ 0.9	53.8 $\pm$ 1.6
AirPhyNet [†]	9.70	3.32	53.7	54.7
AirDualODE	–	–	–	–
AirDDE	–	–	–	–
TCN-DIR	10.30 $\pm$ 0.23	5.04 $\pm$ 0.34	122.9 $\pm$ 11.4	49.7 $\pm$ 1.7
STMamba	<u>9.43</u> $\pm$ 0.11	3.24 $\pm$ 0.05	53.4 $\pm$ 2.4	52.7 $\pm$ 0.9
AirLapseV2	9.35 $\pm$ 0.04	3.19 $\pm$ 0.03	52.0 $\pm$ 1.6	<u>52.0</u> $\pm$ 0.8

Table 9

High-pollution event detection on the test split at the US AQI thresholds (mean $\pm$ std over 5 seeds). CSI and POD: higher is better; FAR: lower is better. **Bold**/underlined: best/second-best learned model for CSI and POD. FAR is not ranked, because a model that rarely forecasts exceedances gets a trivially low FAR. [†]Fewer than five seeds. [‡]The MLP never forecasts an exceedance, so its FAR is undefined.

Model	$\geq 35.5 \mu\text{g m}^{-3}$			$\geq 55.5 \mu\text{g m}^{-3}$		
	CSI	POD	FAR	CSI	POD	FAR
Persistence	0.432	0.604	0.396	0.397	0.569	0.433
MLP	0.000	0.000	n/a [‡]	0.000	0.000	n/a [‡]
LSTM	0.342 $\pm$ 0.022	0.387 $\pm$ 0.031	0.250 $\pm$ 0.020	0.228 $\pm$ 0.040	0.250 $\pm$ 0.051	0.253 $\pm$ 0.046
GRU	0.373 $\pm$ 0.024	0.425 $\pm$ 0.039	0.243 $\pm$ 0.030	0.126 $\pm$ 0.132	0.137 $\pm$ 0.147	0.140 $\pm$ 0.104
Transformer	–	–	–	–	–	–
Informer	0.400 $\pm$ 0.020	0.475 $\pm$ 0.037	0.279 $\pm$ 0.024	0.279 $\pm$ 0.045	0.312 $\pm$ 0.060	0.261 $\pm$ 0.028
Autoformer	0.391 $\pm$ 0.007	0.490 $\pm$ 0.007	0.342 $\pm$ 0.009	<u>0.330</u> $\pm$ 0.007	<u>0.420</u> $\pm$ 0.009	0.394 $\pm$ 0.012
Crossformer	0.296 $\pm$ 0.025	0.327 $\pm$ 0.031	0.247 $\pm$ 0.014	0.114 $\pm$ 0.027	0.121 $\pm$ 0.030	0.307 $\pm$ 0.039
PM _{2.5} -GNN	0.420 $\pm$ 0.008	0.489 $\pm$ 0.012	0.253 $\pm$ 0.005	0.366 $\pm$ 0.009	0.419 $\pm$ 0.016	0.256 $\pm$ 0.014
AirFormer [†]	0.388 $\pm$ 0.033	0.479 $\pm$ 0.058	0.322 $\pm$ 0.023	0.293 $\pm$ 0.043	0.345 $\pm$ 0.077	0.314 $\pm$ 0.059
AirPhyNet [†]	<u>0.415</u>	<u>0.490</u>	0.270	0.299	0.328	0.227
AirDualODE	–	–	–	–	–	–
AirDDE	–	–	–	–	–	–
TCN-DIR	0.397 $\pm$ 0.011	0.600 $\pm$ 0.056	0.456 $\pm$ 0.037	0.325 $\pm$ 0.035	0.440 $\pm$ 0.073	0.438 $\pm$ 0.043
STMamba	0.404 $\pm$ 0.021	0.478 $\pm$ 0.034	0.274 $\pm$ 0.032	0.309 $\pm$ 0.037	0.353 $\pm$ 0.050	0.281 $\pm$ 0.035
AirLapseV2	0.391 $\pm$ 0.033	0.468 $\pm$ 0.063	0.286 $\pm$ 0.043	0.296 $\pm$ 0.039	0.348 $\pm$ 0.062	0.320 $\pm$ 0.050

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